

MATERIAL SAFETY DATA SHEET

1. IDENTIFICATION

A. Product name : Ginsenosides Solution

B. Recommended use and restriction on use

1) **General use :** Not available

2) **Restriction on use :** Not available

C. Manufacturer / Supplier / Distributor information

- Company name : OPTBIO Inc.

- 137-162, Gwahakdanji-ro, Sacheon-myeon, Gangneung-si, Gangwon-do, , Republic of Korea

- Telephone number : +82 33 645 0100

2. HAZARD IDENTIFICATION

A. GHS Classification

- Not applicable

B. GHS label elements

1) **Hazard sysmbols :** Not applicable

2) **Signal words :** Not applicable

3) **Hazard statements :** Not applicable

4) **Precautionary statements**

- Prevention : Not applicable

- Response : Not applicable

- Storage : Not applicable

- Disposal : Not applicable

C. Other hazards which do not result in classification : Not applicable

3. COMPOSITION / INFORMATION ON INGREDIENTS

INCI	CAS No.	Content (%)
Water	7732-18-5	69.9
Butylene Glycol	107-88-0	30.0
Ginsenosides	62025-49-4 52286-59-6	0.1
Total		100.0

4. First AID MEASURES

A. Eye contact

- In case of contact with material, immediately flush eyes with running water for at least 20 minutes.

- Get medical attention immediately.

B. Skin contact

- In case of contact with material, immediately flush skin and eyes with running water for at least 20 minutes.

- Get medical attention immediately.
- Remove and isolate contaminated clothing and shoes.
- Wash clothes and shoes thoroughly before reuse.

C. Inhalation contact

- Get emergency medical attention.
- Remove to fresh air.
- If not breathing, give artificial respiration.
- If breathing is difficult, give oxygen.

D. Ingestion Contact

- Don't feed anything to a unconscious person by mouth.
- Get emergency medical attention.

E. Delayed and immediate effects and chronic effects form short and long term exposure

- Do not administer adrenaline.
- Ensure that medical personnel are aware of the substance and take protective action.

5. FIREFIGHTING MEASURES

A. Suitable (Unsuitable) extinguishing media

- Small fires: dry sand, dry chemical, alcohol-resistant foam, water spray, general foam, CO₂ (suitable extinguishing media).
- Large fires: Water spray/mist, regular foam (suitable extinguishing media).
- High pressure water (unsuitable extinguishing media).

B. Specific hazards arising from the chemical

- The container can explode when heated.
- It can be ignited by heat, sparks, flames.

C. Special protective actions for firefighters

- May be ignited by heat, sparks or flames.
- In case of tank fire, cool containers with plenty of water after extinguishing is extinguished.
- In case of tank fire, step away from tanks in flames.
- In case of tank fire, if there is a loud noise from the pressure relief device or if the tank is discolored, leave immediately.
- Some may be transported at high temperatures.
- Spills may cause contamination.
- Contact may cause skin and eye burns.

6. ACCIDENTAL RELEASE MEASURES

A. Personal precautions, protective equipment and emergency procedures

- Eliminate all ignition sources.
- Stop leak if it is not hazardous.
- Be aware of substances and conditions to avoid.
- Ventilate contaminated area.
- Do not touch or walk on the display.
- Avoid dust formation.
- Do not enter the space without adequate protective equipment, such as a respirator or respirator, until adequate air (oxygen concentration of 18 to 23.5%) is obtained.

B. Environmental precautions

- Prevent entry into waterways, sewers, basements and confined spaces.

C. Methods and materials for containment and cleaning up

- For small spills, flush contaminated area with plenty of water.
- For small spills, absorb with sand, non-combustible material and place in a container.
- In case of large spills, ditch liquid spillage away.
- With a clean shovel, place spillage in a clean, dry container with a loosely closed seal and remove container from spill area.
- In case of powder leakage, cover with plastic sheet to prevent diffusion and keep dry.

7. HANDLING AND STORAGE

A. Precautions for safe handling

- Wash the handling area thoroughly after handling.
- Follow all MSDS/label precautions as product residue may remain after container is emptied.
- Pay attention to incompatible materials and conditions.
- Work with reference to engineering controls and personal protective equipment.
- Beware of high temperature.
- Be careful not to leak as it may cause suffocation in a closed place by lowering the oxygen concentration in the air when the substance is leaked.
- Check the oxygen concentration before entering the area as there is a risk of loss of consciousness or death due to oxygen deficiency in high concentrations in the air.
- When a substance is spilled, be careful not to spill it as there is a risk of serious suffocation when in a closed place as the liquid evaporates quickly and displaces air.
- At 20°C, this material evaporates rather slowly and reaches a hazardous concentration, so keep it below 20°C.
- Evaporation rarely occurs at 20°C, but do not spray as spraying can reach harmful concentrations of air particles very quickly.
- Do not spray or spray as it evaporates faster if sprayed or sprayed.

B. Conditions for safe storage, including any incompatibilities

- Keep tightly closed.
- Store in a cool, dry place.
- Be aware of substances and conditions to avoid.
- Empty drums should be drained completely and properly closed, immediately returned to drum control or properly placed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

A. Domestic exposure standard

- Exposure limits : Not available
- ACGIH TLV : Not available
- Biological exposure criterion : Not available
- Other exposure criteria : Not available

B. Engineering controls

- Use process isolation, local exhaust or keep air levels below exposure standards.
- Install wash and safety shower facilities for storing or using this material.

C. Individual protection measures, such as personal protective equipment

1) Respiratory protection

- Wear respiratory protective equipment certified by the Occupational Safety and Health Service to suit the physical and chemical properties of the gas/liquid being exposed.

- In the case of gas/liquid substances, the following respiratory protection is recommended - Isolation full-face respirator (for organic compounds (for acid gases, in the case of acid gases)) or isolating half-type respirator (for organic compounds (for acid gases)) (for acid gas in case of acid gas)) or direct connection type full-scale gas mask (for organic compounds (for acid gas in case of acid gas)) or half-type gas mask (for organic compound (for acid gas in case of acid gas)) or electric gas mask .
- In case of insufficient oxygen (<19.5%), wear a ventilation mask or self-contained breathing apparatus.

2) Eye protection

- Wear safety glasses or breathable safety goggles to protect your eyes from organic substances in vapor form that can cause eye irritation or other health hazards.
- Install emergency cleaning facilities (shower type) and cleansing facilities in places that are easily accessible to workers.

3) Hand protection

- Wear protective gloves of appropriate material, taking into account the physical and chemical properties of chemicals.

4) Skin Protection

- Wear protective clothing of appropriate material considering the physical and chemical properties of the chemical.

9. PHYSICAL AND CHEMICAL PROPERTIES	
A. Appearance	
- Appearance	Colorless liquid
- Color	Yellow to brown
B. Odour	Typical
C. Odour threshold	Not available
D. pH	3.50 ~ 5.50
E. Purity of Peptides by HPLC	Not available
F. Initial Boiling Point/Boiling	Not available
G. Flash point	Not available
H. Evaporation rate	Not available
I. Flammability (solid, gas)	Not available
J. Upper/Lower Flammability or explosive limits	Not available
K. Vapour pressure	Not available
L. Solubility	Not available
M. Vapour density	Not available
N. Specific gravity (Relative density)	0.990 ~ 1.030
O. Partition coefficient of n-octanol/water	Not available
P. Autoignition temperature	Not available
Q. Decomposition temperature	Not available
R. Viscosity	Not available
S. Molecular weight	Not available

10. STABILITY AND REACTIVITY

A. Chemical Stability

- Container may explode when heated.
- Some can burn but do not ignite easily.
- Non-flammable, the material itself does not burn, but it may decompose when heated to generate corrosive/toxic fumes.
- May generate irritating, corrosive and toxic gases in case of fire.
- Stable under normal temperature and pressure conditions.
- Inhalation of material may be harmful.
- Some liquids may produce vapors that cause vertigo and suffocation.

B. Conditions to avoid

- Heat, sparks, flames, fire source.

C. Materials to avoid

- Combustible material, Reductive Substances.
- Irritating, toxic gas.

D. Harmful substances produced in decomposition

- Not applicable.

11. TOXICOLOGICAL INFORMATION

A. Information on the likely routes of exposure

- Can be absorbed by the body by inhalation
- Can be absorbed by the body by inhalation and digestive
- Can be absorbed by the body through the skin, through the digestive tract, or by inhalation of an aerosol
- Can be absorbed by the body by inhalation of steam

B. Delayed and immediate effects and chronic effects from short and long term exposure

1) Acute toxicity

- Oral toxicity

[Butylene Glycol] : LD50 22800 mg/kg Test species: Rat (Route of administration: gastric tube, male) ※Source: ECHA

- Dermal toxicity

[Butylene Glycol] : LD50 >20000 mg/kg Test species: Rabbit ※Source: ECHA

- Inhalation toxicity

[Butylene Glycol] : LC0 78 ppm 8 hr Test species: Rat ※Source: ECHA

2) Skin corrosion/irritation

[Butylene Glycol] : Research that cannot be used for classification ※Source: ECHA

3) Serious eye damage/irritation

[Butylene Glycol] : Slightly irritating, Rabbit ※Source : ECHA

4) Respiratory sensitization

[Butylene Glycol] : Not applicable.

5) Skin sensitization

[Butylene Glycol] : Human body/non-response ※Source: IUCLID

6) Carcinogenicity

- * IARC : Not applicable.
- * OSHA : Not applicable.
- * ACGIH : Not applicable.
- * NTP : Not applicable.
- * EU CLP : Not applicable.

7) Germ cell mutagenicity

[Butylene Glycol] : in vivo -Chromosomal aberration test using mammalian bone marrow cells: negative (rat, male/female) in vitro -Bacterial reverse mutation test: negative (S. typhimurium TA1535, regardless of metabolic activity), OECD TG 471, OECD TG 472, GLP ※Source : ECHA

8) Reproductive toxicity

[Butylene Glycol] : 1,3-butylene glycol did not affect fertility in a fifth-generation study at concentrations up to 10% from dietary intake (5000 mg/kg bw/d). No second generation was produced in the fifth pups at the highest concentration tested (24%, 12000 mg/kg bw/d). No teratogenic effects in rats treated with 1,3-butylene glycol up to 24% (12000 mg/kg bw/d) in diet, 10% (5000 mg/kg bw/d) in diet 1,3-butylene Fetal toxicity effects occurred at concentrations higher than glycol, rat) ※Source: ECHA

9) STOT-single exposure

[Butylene Glycol] : Inhalation- No lethal effect was observed in male rats after a single 8-hour exposure to saturated vapor (concentration was about 78 ppm or 292 mg/m³) (Smyth et al., 1951) ※Source: ECHA

10) STOT-repeated exposure

[Butylene Glycol] : Oral (Chronic)- No test-related side effects were observed in a chronic lactation study of rats administered with 10% (5000 mg/kg/d) 1,3-butylene glycol in the diet, Rat): 20100 mg/kg bw/day of 20100 mg/kg bw/day for 2 hours/day for 14 or 4 days, respectively, on the intact or wounded skin of guinea pigs, no side effects were observed. , Guinea pig ※Source: ECHA

11) Aspiration hazard : Not available

12) Other Hazard Effects : Not available

12. ECOLOGICAL INFORMATION

A. Ecotoxicity

1) Pisces

[Butylene Glycol] : LC50 > 100 mg/l 96 hr Oryzias latipes (OECD TG 203)
※Source : ECHA

2) Shellfish

[Butylene Glycol] : EC50 > 1000 mg/l 48 hr Daphnia magna (OECD TG 202 , fresh water, GLP) ※
Source : ECHA

3) Birds

[Butylene Glycol] : NOEC ≥ 1070 mg/l 72 hr Desmodemus subspicatus (OECD TG 201, EU Method C.3 (Algal Inhibition test), fresh water, GLP) ※Source : ECHA

B. Persistence and degradability

1) Persistence

[Butylene Glycol] : 0.13 (Pow, 25°C) ※Source : ECHA

2) Degradability

[Butylene Glycol] : Not available

C. Bioaccumulative

1) Bioaccumulative

[Butylene Glycol] : 3.2 BCF (l/kg) ※Source : ECHA

2) Biodegradable

[Butylene Glycol] : 81 (%) 29 day (CO₂ evolution) ※Source : ECHA

D. Mobility in soil : Not available

E. Other adverse effects : Not available

13. DISPOSAL CONSIDERATIONS

A. Disposal methods

- Dispose of contents and container in accordance with regulations if specified in the Waste Management Act.

B. Special precautions for disposal

- Not applicable

14. TRANSPORT INFORMATION

A. UN No. (IMDG CODE/IATA DGR) : Not applicable

B. UN Proper Shipping Name : Not applicable

C. Risk grade in transportation : Not applicable

D. IMDG CODE/IATA DGR Packing group : Not applicable

E. Marine pollutant : Not applicable

F. Special precautions for user related to transport or transportation measures

- Type of emergency measures in case of fire : Not applicable
- Type of emergency measures in case of leak : Not applicable

15. REGULATORY INFORMATION

A. Regulations under the Occupational Safety and Health Act : Not applicable

B. Act on Registration and Evaluation of Chemical Substances

[Butylene Glycol] : Not applicable

C. Regulations under the Dangerous Substances Safety Management Act

[Butylene Glycol] : Class 4, tertiary petroleum (water soluble) (4000 t)

D. Regulations by the Waste Management Act : Not applicable

E. Regulations by other domestic and foreign laws

1) Korea

- Not applicable

2) United States

- * OSHA PROCESS SAFETY (29CFR1910.119) : Not applicable
- * CERCLA Section 103 (40CFR302.4) : Not applicable
- * EPCRA Section 302 (40CFR355.30) : Not applicable
- * EPCRA Section 304 (40CFR355.40) : Not applicable
- * EPCRA Section 313 (40CFR372.65) : Not applicable
- * Rotterdam Convention listed ingredients : Not applicable
- * Stockholm Convention listed ingredients : Not applicable
- * Montreal Protocol material : Not applicable

16. OTHER INFORMATION

A. Reference

- This MSDS is prepared in consideration of the current status of domestic regulations and regulations based on the Ministry of Employment and Labor Notice No. 2020-130 (standards for classification, labeling and material safety and health data).
- This Safety Data Sheet was compiled with data and information from the following sources: KOSHA, ECHA, GESTIS, HSDB

B. Issue date

- 2023-02-17

C. Revision number and Last date revised

- 0 times

D. Other

- Not applicable